PAPER_503: UQFF Full Lagrangian Wolfram Syntax Export

Unified Quantum Field Framework (UQFF)

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Source: PAPER_503_UQFF_Lagrangian_Wolfram_Export.md

PAPER_503: UQFF Full Lagrangian Wolfram Syntax Export

Session: 137 | **Source:** grok_share_84a767d3.txt (lines 700–1100) **Date:** November 2025 **Related files:** source175_uqff_wolfram_export.cpp

§1.1 Abstract

The UQFF Full Lagrangian Wolfram Export assembles a complete 26-dimensional unified field Lagrangian from the 8+ source-file contributions and sends it to the embedded WSTP kernel for symbolic reduction via `FullSimplify`. This paper documents the Lagrangian structure, the export protocol, the precision settings, and the expected simplification outcomes.

§1.2 UQFF Master Lagrangian (Wolfram Syntax)

The full Lagrangian `masterUQFF` is defined as the sum of the following sectors:

```
masterUQFF = (c^4/(8*Pi*G)) * RicciScalar      (* GR gravity sector *)
           - (1/4) * F_uv * F^uv               (* Maxwell electroweak EM *)
           + I * Psi_bar * GammaMu * D_Mu * Psi - m_f * Psi_bar * Psi  (* Dirac fermion *)
           + D_Mu*Phi * ConjugateTranspose[D^Mu*Phi] - lambda_H*(Phi^2 - v_H^2/2)^2  (* Higgs *)
           + alpha_GB * (Riem^2 - 4*Ric^2 + R^2)   (* Gauss-Bonnet higher curvature *)
           + (1/V_22D) * Integral22D[L_internal]    (* 22D Calabi-Yau compactification *)
           + StarMagicAetherFlow                    (* UQFF aether field term *)
           + StarMagicHypergraphRuleEmbedding        (* Wolfram hypergraph layer *)
           + SOURCE171_8Systems                     (* 8 astrophysical systems, source171 *)
           + SOURCE172_19Systems_26D                (* 19-system 26D unification, source172 *)
           + SOURCE173_HypergraphLayer              (* Hypergraph causal graph, source173 *)
```

§1.3 Precision Settings

```
$MaxExtraPrecision = 20000;   (* Allow up to 60,000-bit intermediate arithmetic *)
$MinPrecision = 50;           (* Minimum 50-decimal-place guarantee *)
```

The `ComplexityFunction` passed to `FullSimplify` ranks expressions by:

1. Total symbol count (favors shorter)
2. Presence of contracted indices (favors tensors over components)
3. Presence of UQFF calibrated constants κ, [SSq], H_SCm

§1.4 Export Protocol

```
Step 1: InitializeWolframKernel() → persistent kernel
Step 2: Set precision:
        WolframEvalToString(&quot;$MaxExtraPrecision = 20000;&quot;)
Step 3: Define constants:
        WolframEvalToString(&quot;kappa=0.0005; SSq=0.57; H_SCm=0.99; U_UA=1e-4;&quot;)
Step 4: Define and send masterUQFF symbol (string ≤ 100 MB):
        WolframEvalToString(lagrangian_definition_string)
Step 5: Simplify:
        WolframEvalToString(&quot;ToString[FullSimplify[masterUQFF, ComplexFunction-&gt;...], InputForm]&
Step 6: Print verdict
```

§1.5 Source Term Inventory

Source File	Systems	Terms
source171.cpp	8 astrophysical (SGR1745, SgrA*, Tapestry, Westerlund, Pillars, Rings, Student)	18-term MUGE
source172.cpp	19-system 26D master framework	26-layer polynomial equations
source173.cpp	Wolfram hypergraph replacement layer	Causal graph rules
source174.cpp	WSTP bridge itself	Connection monitoring term

§1.6 Simplification Outcome Classes

Outcome	Meaning	Action
Result = 0	Full cancellation — field theory closes	Publish: UQFF proven
Result = small constant	Near-unification with calibration needed	Adjust κ, [SSq], H_SCm
Result = reduced tensor	Partial simplification	Expand remaining source files
Timeout / memory OOM	Expression too large for current terms	Split by sector, simplify each

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§1.7 Calibrated UQFF Constants

From UQFF 99.9% solvability analysis (Grok 4, Sept 14–21, 2025):

```
κ          = 0.0005 / day     (Hubble-rate decay coupling)
[SSq]      = 0.57             (superconductive square bracket)
H_SCm      ≈ 0.99             (superconductor magnetic suppression)
U_UA       ≈ 0.0001           (undefined aether coupling)
k_η        = 10^{-113}        (vacuum coupling)
β_i        ≈ 0.603            (buoyancy amplitude)
```

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§1.8 Citation

Source: grok_share_84a767d3.txt, lines 700–1100 Related: source175_uqff_wolfram_export.cpp Paper number: PAPER_503